

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			award (1) for any of following • H_2SO_4 loses protons / H^+ ions and NH_3 gains protons / H^+ ions • H_2SO_4 is a proton donor and NH_3 is a proton acceptor • NH_3 accepts a proton from H_2SO_4 neutral answer – reference to salt formation			1	1		
	(b)			$\text{moles } (\text{NH}_4)_2\text{SO}_4 = \frac{1.86}{132.18} = 0.0141 \quad (1)$ $\text{moles NaOH} = 0.0282 \quad (1)$ $\text{concentration} = \frac{0.0282}{0.0267} = 1.06 \text{ mol dm}^{-3} \quad (1)$		3		3	1 1	
	(c)	(i)		if a system at equilibrium is subjected to a change then the position of equilibrium will shift to minimise/oppose that change	1			1		
		(ii)		student correct since more moles of ammonia present at equilibrium at higher temperature (1) system opposes increase in temperature by shifting equilibrium position to the left / in the endothermic direction (1) neutral answer – reference to the ‘endothermic <u>side</u> ’ if incorrect award no marks accept converse answer student correct since fewer moles of ammonia present at equilibrium at lower temperature (1) system opposes decrease in temperature by shifting equilibrium position to the right / in the exothermic direction (1)	1		1	2		

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		(iii)		(1) catalyst does not affect the position of equilibrium / number of moles present at equilibrium but equilibrium reached more quickly / reaction is faster (1)	1	1		2		
	(d)			1.26 / 1.3		1		1	1	
	(e)			nitrogen more electronegative than hydrogen / nitrogen and hydrogen have different electronegativities (1) this results in polarity / unequal electron distribution in the bond (1)	1		1	2		
				Question 9 total	4	5	3	12	3	0